

Examples: Hypothesis Testing

STAT 521 – Applied Probability and Statistical Inference

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Example 1: Aerobics Program

Average heart rate for Americans is 72 beats/minute. A group of 25 individuals participated in an aerobics fitness program to lower their heart rate. After six months the group was evaluated to identify if the program had significantly slowed their heart. The mean heart rate for the group was 69 beats/minute with a standard deviation of 6.5. Was the aerobics program effective in lowering heart rate?

Example 1: Aerobics Program (Setup)

Context: Average resting heart rate in the population is 72 bpm. After a 6-month aerobics program, a sample of 25 participants has:

$$\bar{x} = 69, \quad s = 6.5.$$

Goal: Test if the program *lowered* mean heart rate.

Parameter:

μ = true mean resting heart rate after the program.

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Hypotheses (left-tailed test):

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Conditions:

- Sample is (approximately) a simple random sample.
- Population of heart rates roughly normal or no strong skew with $n = 25$.
- σ unknown

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Conditions:

- Sample is (approximately) a simple random sample.
- Population of heart rates roughly normal or no strong skew with $n = 25$.
- σ unknown

→ Use a ***t*-test**.

Example 1: Aerobics Program (Calculation)

Test statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{69 - 72}{6.5/\sqrt{25}} = \frac{-3}{6.5/5} \approx -2.31.$$

Degrees of freedom: $df = n - 1 = 24$.

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Degrees of freedom: $df = n - 1 = 24$.

p-value (left-tailed):

$$p = P(T_{24} < -2.31) \approx 0.015.$$

Example 1: Aerobics Program (Conclusion)

Decision at $\alpha = 0.05$:

$$p \approx 0.015 < 0.05 \quad \Rightarrow \quad \text{Reject } H_0.$$

Conclusion: There is statistically significant evidence that the program *reduced* the mean resting heart rate below 72 bpm.

Example 2: Shaq's Free Throws

Any basketball fan knows that Shaquille O'Neal, one of the NBA's most dominant centers of the last twenty years, always had difficulty shooting free throws. Over the course of his career, his overall made free throw percentage was 53.3%. During one offseason, Shaq had been working with an assistant coach on his free-throw technique. During the next season, a simple random sample showed that Shaq made 26 of 39 free-throw attempts. Test the claim at the 0.05 SL that Shaq has significantly improved his free-throw shooting.

Example 2: Shaq's Free Throws (Setup)

Context: Over his career, Shaq's free-throw percentage was 53.3%. After working on his technique, in a sample of 39 attempts he made 26 shots.

Goal: Test the claim that Shaq has significantly improved his free-throw shooting at 5% significance level.

$$p_0 = 0.533, \quad n = 39, \quad x = 26, \quad \hat{p} = \frac{26}{39} \approx 0.667.$$

Parameter:

p = true free-throw success probability after training.

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$$p_0 = 0.533, \quad n = 39, \quad x = 26, \quad \hat{p} = \frac{26}{39} \approx 0.667.$$

Parameter:

p = true free-throw success probability after training.

Hypotheses (right-tailed):

$$H_0 : p = 0.533 \quad \text{vs} \quad H_a : p > 0.533.$$

Example 2: Shaq's Free Throws (Calculation)

Check conditions for normal approx:

$$np_0 \approx 20.8 \geq 5, \quad n(1 - p_0) \approx 18.2 \geq 5.$$

→ A **z-test** is appropriate.

Test statistic:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}} \approx \frac{0.667 - 0.533}{\sqrt{\frac{0.533 \cdot 0.467}{39}}} \approx 1.67.$$

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p-value (right-tailed):

$$p = P(Z > 1.67) \approx 0.047.$$

Example 2: Shaq's Free Throws (Conclusion)

Decision at $\alpha = 0.05$:

$$p \approx 0.047 < 0.05 \quad \Rightarrow \quad \text{Reject } H_0.$$

Conclusion (in context): There is **statistically significant evidence** (at the 5% level) that Shaq's true free-throw percentage improved.

Example 3: GRE Scores

The population of all verbal GRE scores are known to have a standard deviation of 8.5. The UW Psychology department hopes to receive applicants with a verbal GRE score over 210. This year, the mean verbal GRE scores for the 42 applicants was 212.79. Using a value of $\alpha = 0.05$ is this new mean significantly greater than the desired mean of 210?

Example 3: GRE Scores (Setup)

Context: Population SD of verbal GRE is known to be $\sigma = 8.5$. The department hopes for a mean score above 210. This year:

Goal: Test if the mean verbal GRE score is greater than the target GRE score.

$$n = 42, \quad \bar{x} = 212.79, \quad \sigma = 8.5.$$

Parameter:

μ = true mean verbal GRE score of this year's applicants.

Example 3: GRE Scores (Setup)

Context: Population SD of verbal GRE is known to be $\sigma = 8.5$. The department hopes for a mean score above 210. This year:

Goal: Test if the mean verbal GRE score is greater than the target GRE score.

$$n = 42, \quad \bar{x} = 212.79, \quad \sigma = 8.5.$$

Parameter:

μ = true mean verbal GRE score of this year's applicants.

Hypotheses (right-tailed):

$$H_0 : \mu = 210 \quad \text{vs} \quad H_a : \mu > 210.$$

Example 3: GRE Scores (Calculation)

Conditions:

- SRS of applicants.
- Either scores are approx. normal or n is large $\Rightarrow$ CLT.
- σ known .

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$\Rightarrow$ use a **z-test for the mean**

Test statistic:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{212.79 - 210}{8.5/\sqrt{42}} \approx \frac{2.79}{1.31} \approx 2.13.$$

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- SRS of applicants.
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- σ known .

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Test statistic:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{212.79 - 210}{8.5/\sqrt{42}} \approx \frac{2.79}{1.31} \approx 2.13.$$

p-value (right-tailed):

$$p = P(Z > 2.13) \approx 0.017.$$

Example 3: GRE Scores (Conclusion)

Decision at $\alpha = 0.05$:

$$p \approx 0.017 < 0.05 \quad \Rightarrow \quad \text{Reject } H_0.$$

Conclusion (in context): There is **statistically significant evidence** that the true mean verbal GRE score of this year's applicants is *greater than* 210.